4. Kalai wants to cut identical squares as big as she can, from a piece of paper measuring 168mm and by 196mm. What is the length of the side of the biggest square? (To find HCF using repeated subtraction method)

## Objective Type Questions

5. What is the eleventh Fibonacci number?
   (a) 55 (b) 77 (c) 89 (d) 144

6. If F(n) is a Fibonacci number and n =8, which of the following is true?
   (a) F(8) = F(9)+F(10) (b) F(8) = F(7)+F(6) (c) F(8) = F(10)×F(9) (d) F(8) = F(7)–F(6)

7. Every 3rd number of the Fibonacci sequence is a multiple of _____
   (a) 2 (b) 3 (c) 5 (d) 8

8. Every _____ number of the Fibonacci sequence is a multiple of 8
   (a) $2^{nd}$ (b) $4^{th}$ (c) $6^{th}$ (d) $8^{th}$

9. The difference between the $18^{th}$ and $17^{th}$ Fibonacci number is
   (a) 233 (b) 377 (c) 610 (d) 987

10. Common prime factors of 30 and 250 are
    (a) 2 x 5 (b) 3 x 5 (c) 2 x 3 x 5 (d) 5 x 5

11. Common prime factors of 36, 60 and 72 are
    (a) 2 x 2 (b) 2 x 3 (c) 3 x 3 (d) 3 x 2 x 2

12. Two numbers are said to be co-prime numbers if their HCF is
    (a) 2 (b) 3 (c) 0 (d) 1

## 7.7 Cryptology

In today's world, security in information is a fundamental necessity not only for military and political departments but also for private communication. Today's world of communication has increased the importance of financial data exchange, image processing, biometrics and e-commerce transaction which in turn has made data security an important issue. Cryptology is defined as the science which is concerned with communication in secured form.

### 7.7.1 Cryptology – Some technical details

**Plain text:** The original message is called plain text.

**Cipher text or Cipher number:** The encrypted output (converted message into code) is called Cipher text or Cipher number. Cipher text is written in capital letters, while plain text is usually written in lowercase. A secret key is to use something to generate the Cipher text from the plain text.

**Encryption and Decryption:** The process of converting the plain text to the Cipher text is called encryption and the vice versa is called decryption.



8th_Maths_English Medium_Chapter 7.indd 254 03-12-2022 15:22:06